From VCF to RDF: RML-Based Conversion Approach for the Semantic Representation of Variant Data

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Problem

Limited Query Capabilities
- Requires **specialized tools**
- Not flexible or scalable

Structural Complexity of VCF
- Highly **heterogeneous and irregular fields**:
  - ➢ INFO, FORMAT, GENOTYPE, HEADER

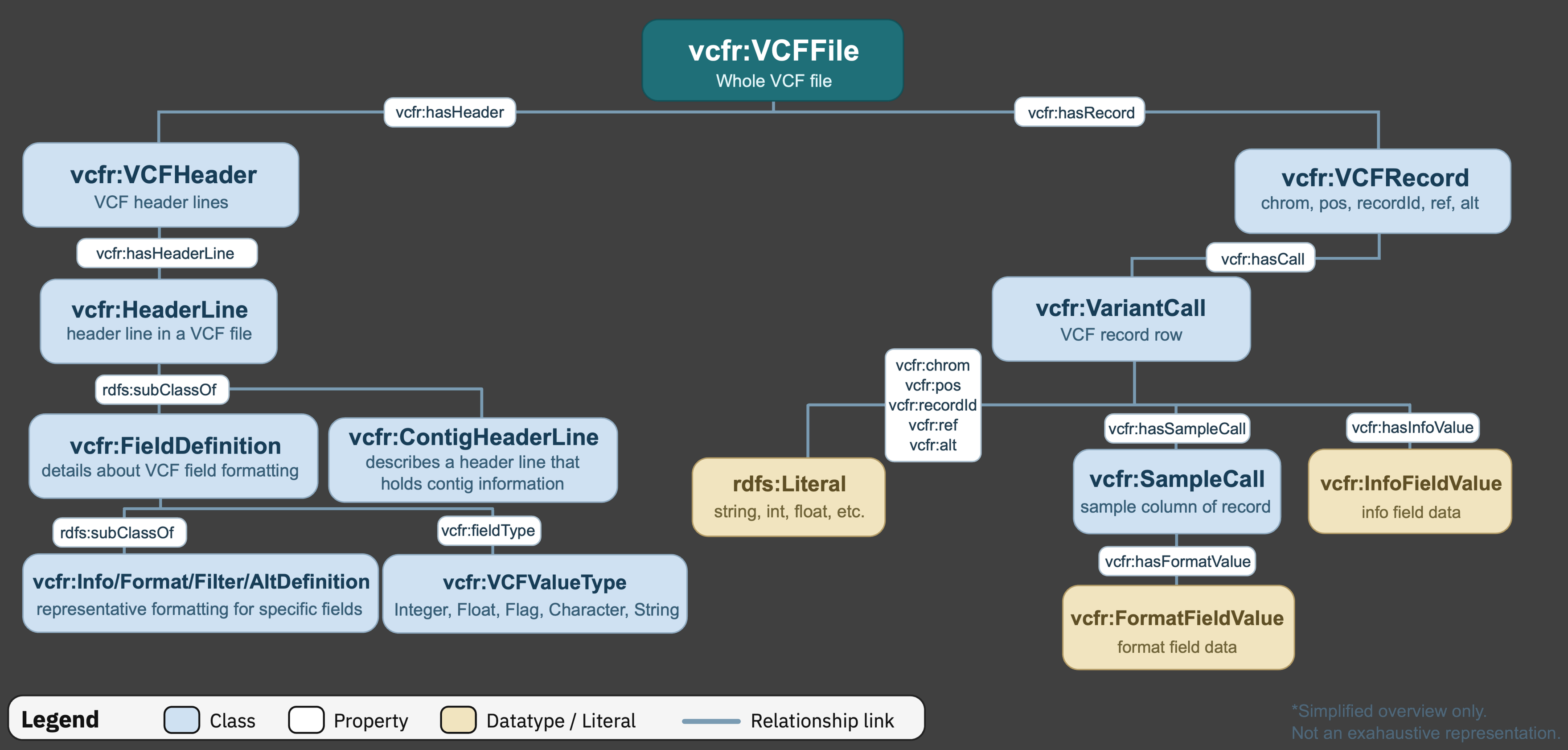
Limited Interoperability
- Difficult to integrate with other biomedical datasets

Challenges

Weak Data Linking & Semantics
- Only **inconsistent / weak references** (e.g., rsIDs)
- No rich semantic structure

Poor Data Access & Privacy Control
- All-or-nothing data sharing

VCF Vocabulary



Semantic VCF Vocabulary

https://w3id.org/vcf-rdfizer/vocab#

1. Full VCF-to-RDF Representation
- Models the **entire VCF file**

2. Focus on Provenance & Structure
- Explicitly captures **HEADERS, row-level data, and call-level provenance**

3. Interoperability via Ontology Alignment
- links to **existing genomic ontologies** (e.g., SO, FALDO, GENO, etc.)

RML Rule-Set

RML for VCF Transformation

1. Declarative & Reusable Mappings
- Define transformations as rules, not code.

2. Handles VCF complexity
- Supports nested, irregular fields (INFO, FORMAT, GENOTYPE).

3. Semantic interoperability
- Enables mapping to ontologies for FAIR, linked data.

4. Scales to large datasets
- Streaming approach via RMLStreamer

rml_rules.ttl

```
<#VCFRecordMap>
  a rr:TriplesMap ;
  rml:logicalSource [
    rml:source [
      a csvw:Table ;
      csvw:url "/data/tsv/records.tsv" ;
      csvw:dialect [
        a csvw:Dialect ;
        csvw:delimiter "\t" ;
        csvw:headerRowCount 1
      ]
    ] ;
    rml:referenceFormulation ql:CSV
  ] ;
  rr:subjectMap [
    rr:template "file://{SOURCE_FILE}#record/{ROW_ID}" ;
    rr:class vcfr:VCFRecord
  ] ;
  rr:predicateObjectMap [
    rr:predicate vcfr:chrom ;
    rr:objectMap [ rml:reference "CHROM" ]
  ] ;
  ...
```

\+ *RMLStreamer Engine [1]*

[1] Gerald Haesendonck, et al. RMLio/RMLStreamer: V2.5.0. v2.5.0, Zenodo, 2 June 2023, https://doi.org/10.5281/zenodo.7998156.

RDF Conversion

VCF to RDF Example

example.vcf

```
##fileformat=VCFv4.2
##fileDate=20200615
##source=CLC Genomics Grid Worker 20.9.9 Beta 1 build 209900
##reference=/CLC_References/homo_sapiens/sequence/hg38_no_alt_analysis_set/Homo_sapiens_sequence_hg38_no_alt_analysis_set
##contig=<ID=chr1,length=248956422>
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Total number of filtered reads used to call variants at this position in this sample.">
##FORMAT=<ID=AD,Number=R,Type=Integer,Description="Allele depth, number of filtered reads supporting the alleles, ordered as listed in REF and ALT.">

#CHROM POS ID  REF ALT QUAL  FILTER INFO    FORMAT SAMPLE1    SAMPLE2
chr1   123456 rs123  A   T   52.3   PASS   DP=42;AF=0.2 GT:DP:AD  0/1:42:30,12 0/1:6:3,2  ] ;
```

example.ttl#Header

```
@base <file://> .
@prefix vcfr: <https://w3id.org/vcf-rdfizer/vocab#> .
@prefix xsd:  <http://www.w3.org/2001/XMLSchema#> .

<example-file1.vcf>
  a vcfr:VCFFile ;
  vcfr:fileFormat "VCFv4.2" ;
  vcfr:fileDate "2020-06-15"^^xsd:date ;
  vcfr:sourceSoftware "CLC Genomics Grid Worker 20.9.9 Beta 1 build 209900" ;
  vcfr:referenceGenome "/CLC_References/homo_sapiens/sequence/hg38_no_alt_analysis_set/Homo_sapiens_sequence_hg38_no_alt_analysis_set" ;
  vcfr:contigCount 1 ;
  vcfr:hasHeader <example-file1.vcf#header> ;
  vcfr:hasRecord <example-file1.vcf#record/var1> .

<example-file1.vcf#header/line/1>
  a vcfr:FileFormatHeaderLine ;
  vcfr:headerKey "fileformat" ;
  vcfr:headerValue "VCFv4.2" .

<example-file1.vcf#header/line/2>
  a vcfr:FileDateHeaderLine ;
  vcfr:headerKey "fileDate" ;
  vcfr:headerValue "20200615" .
  ...
```

example.ttl#Record

```
@base <file://> .
@prefix vcfr: <https://w3id.org/vcf-rdfizer/vocab#> .

<example-file1.vcf>
  a vcfr:VCFFile ;
  vcfr:fileFormat "VCFv4.2" ;
  vcfr:hasRecord <example-file1.vcf#record/var1> .

<example-file1.vcf#record/var1>
  a vcfr:VCFRecord ;
  vcfr:chrom "chr1" ;
  vcfr:chromosome <example-file1.vcf#header/line/5> ;
  vcfr:pos 123456 ;
  vcfr:recordId "rs123" ;
  vcfr:ref "A" ;
  vcfr:alt "T" ;
  vcfr:hasCall <example-file1.vcf#call/var1> .

<example-file1.vcf#call/var1>
  a vcfr:VariantCall ;
  vcfr:qual 52.3 ;
  vcfr:filter "PASS" ;
  vcfr:infoRaw "DP=42;AF=0.2" ;
  vcfr:formatRaw "GT:DP:AD" ;
  vcfr:hasInfoValue
    <example-file1.vcf#call/var1/info/AF>,
    <example-file1.vcf#call/var1/info/DP> ;
  ...
```

Future Work:
- Testing of RML Mappings + Conversion Outputs
- Vocabulary Improvements

Contact

Paper

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